① $S_t = S_0 e^{\mu t - \frac{\sigma^2}{2}t + \sigma W_t} \rightarrow dS_t = \mu S_t dt + \sigma S_t dW_t$

$f(t,x)$

$$\boxed{df(t,S_t) = \frac{\partial}{\partial t} f(t,S_t)dt + \frac{\partial}{\partial x} f(t,S_t)dS_t + \frac{1}{2}\frac{\partial^2}{\partial x^2} f(t,S_t)(dS_t)^2}$$

$f(t,x) = S_0 e^{\mu t - \frac{\sigma^2}{2}t + \sigma x}$

$df(t,W_t) = S_t\left(\mu - \frac{\sigma^2}{2}\right)dt + \sigma S_t dW_t + \frac{1}{2}\sigma^2 S_t \cdot dt = \mu S_t dt + \sigma S_t dW_t$

$\frac{\partial}{\partial t} f(t,x) = S_0 e^{\mu t - \frac{\sigma^2}{2}t + \sigma W_t} \cdot \left(\mu - \frac{\sigma^2}{2}\right) = S_t\left(\mu - \frac{\sigma^2}{2}\right)$

$\frac{\partial}{\partial x} f(t,x) = S_0 e^{\mu t - \frac{\sigma^2}{2}t + \sigma W_t} \cdot \sigma = \sigma S_t$

$\frac{\partial}{\partial x} f(t,x) = S_0 e^{\mu t - \frac{\sigma^2}{2}t + \sigma W_t} \cdot \sigma^2 = \sigma^2 S_t$

② $P(S_{2t} > 2S_t) = ?$

$\cancel{S_0} e^{2\mu t - \sigma^2 t + \sigma W_{2t}} > 2\cancel{S_0} e^{\mu t - \frac{\sigma^2}{2}t + \sigma W_t}$ $\quad \cancel{\Big| : \frac{e^{\sigma W_t}}{e^{2\mu t - \sigma^2 t}}}$

$2\mu t - \sigma^2 t + \sigma W_{2t} > \ln 2 + \mu t - \frac{\sigma^2}{2}t + \sigma W_t$

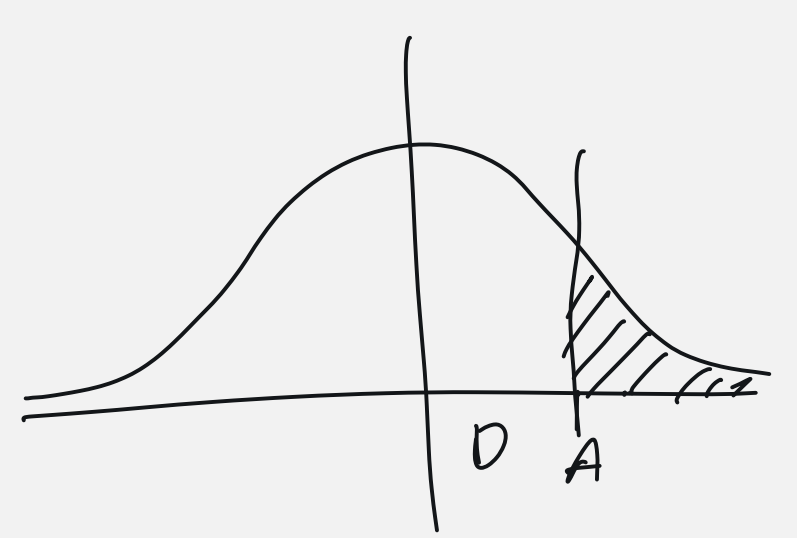
$\sigma(W_{2t} - W_t) > \ln 2 - \mu t + \frac{\sigma^2}{2}t$

$\underbrace{W_{2t} - W_t}_{\sim N(0,t)} > \frac{1}{\sigma}\left(\ln 2 - \mu t + \frac{\sigma^2}{2}t\right)$ $\qquad W_t - W_s \sim N(0, t-s$

$\sqrt{t}X > \frac{1}{\sigma}\left(\ln 2 - \mu t + \frac{\sigma^2}{2}t\right)$

$X \sim N(0,1)$

$X > \underbrace{\frac{1}{\sigma\sqrt{t}}\left(\ln 2 - \mu t + \frac{\sigma^2}{2}t\right)}_{A}$



$P(S_{2t} > 2S_t) = 1 - N(A)$ $\qquad 1 - cdf(A)$

где N = cdf для $N(0,1)$

③ Найти $Var(S_t)$

$Var(X) = E(X^2) - (E(X))^2$

$E(S_t) = E\left(S_0 e^{\mu t - \frac{\sigma^2}{2}t + \sigma W_t}\right) = S_0 e^{\mu t - \frac{\sigma^2}{2}t} E\left(e^{\sigma W_t}\right) =$

$$\boxed{E(e^X) = e^{\frac{1}{2}Var(X)}}$$

$= S_0 e^{\mu t - \frac{\sigma^2}{2}t} e^{\frac{1}{2}Var(\sigma W_t)} = S_0 e^{\mu t \cancel{- \frac{\sigma^2}{2}t}} e^{\cancel{\frac{1}{2}\sigma^2 t}} = S_0 e^{\mu t}$

$Var(X) = E(X - E(X))^2$

$Var(S_t) = E\left(S_t - S_0 e^{\mu t}\right)^2 = E\left[S_0 e^{\mu t}\left(e^{-\frac{\sigma^2}{2}t + \sigma W_t} - 1\right)\right]^2 =$

$= E\left[S_0^2 e^{2\mu t}\left(e^{-\sigma^2 t + 2\sigma W_t} - 2e^{-\frac{\sigma^2}{2}t + \sigma W_t} + 1\right)\right] =$

$= S_0^2 e^{2\mu t}\left(e^{-\sigma^2 t + \frac{1}{2}4\sigma^2 t} - 2e^{\cancel{-\frac{\sigma^2}{2}t + \frac{1}{2}\sigma^2 t}} + 1\right)\Big] =$

$= S_0 e^{2\mu t}\left(e^{\sigma^2 t} - 1\right)$

④ $dS_t = \mu S_t dt + \sigma dW_t$

$P(S_t < 0)$

$S_t = S_0 e^{\mu t} + \int_0^t \sigma e^{\mu(t-s)} dW_s$

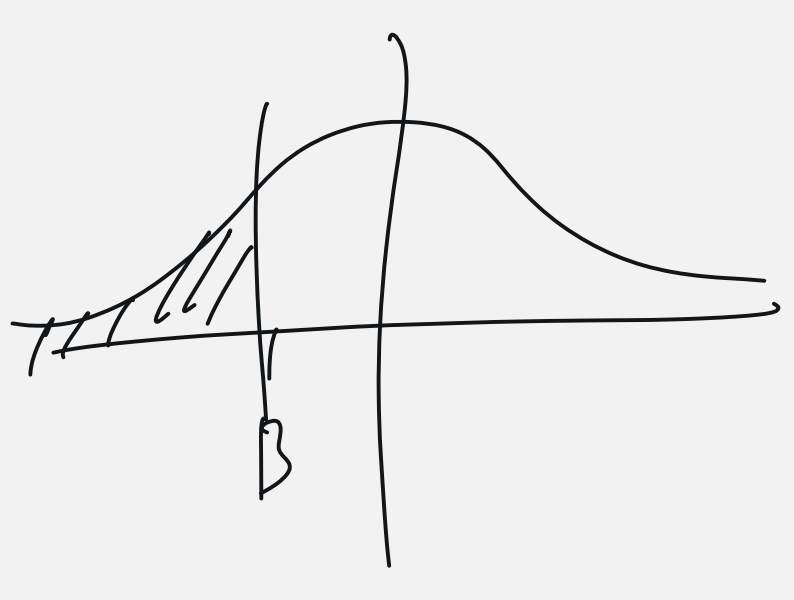
$P(S_t < 0) = P\left(\int_0^t \sigma e^{\mu(t-s)} dW_s < S_0 e^{\mu t}\right)$

$\sim N\left(0, \int_0^t \sigma^2 e^{2\mu(t-s)} ds\right)$

$= \sigma^2 e^{2\mu t} \int_0^t e^{-2\mu s} ds =$

$= \sigma^2 e^{2\mu t}\left(-\frac{1}{2\mu} e^{-2\mu s}\Big|_0^t\right) =$

$= -\frac{\sigma^2 e^{2\mu t}}{2\mu}\left(e^{-2\mu t} - 1\right) = \frac{\sigma^2}{2\mu}\left(e^{2\mu t} - 1\right)$



$P\left(\sqrt{\frac{\sigma^2}{2\mu}(e^{2\mu t} - 1)}\, X < -S_0 e^{\mu t}\right)$

$X \sim N(0,1)$

$P\left(X < \overbrace{-\frac{S_0 e^{\mu t}}{\sqrt{\frac{\sigma^2}{2\mu}(e^{2\mu t} - 1)}}}^{B}\right) = N(B)$

N – cdf норм. распр

⑤ $dB_t = g(t)B_t dt$

$g(t) = r$ $\qquad$ r – risk free

$dA_t = rA_t dt$ → безрисковое вложение

Пусть $A_0 = 1$ и $B_0 = 1$

Построим портфолио

$x(t)$ → кол-во акций

$y(t)$ → деньги в банке

$V_t = x(t)B_t + y(t)A_t$

Выберем $x(t) = \begin{cases} 1, & \text{если } B_t > A_t \\ 0, & \text{если } B_t = A_t \\ -1, & \text{если } B_t < A_t \end{cases}$, $y(t) = -x(t)$

$V_t = \begin{cases} B_t - A_t, & \text{если } B_t > A_t \\ 0, & \text{если } B_t = A_t \\ A_t - B_t, & \text{если } B_t < A_t \end{cases}$

$dV_t = x(t)dB_t + y(t)dA_t$

$V_t = V_0 \quad \forall t$

$V_0 = 0$ по условию отсутствия арбитража

$\Rightarrow V_t \geq 0 \Rightarrow B_t = A_t = e^{rt} \Rightarrow g(t) = r$ почти наверное